

# MATH 1070Q: Mathematics for Business and Economics

## Lecture Slides

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## Topic 3: Percents and Discounts, Markups, Taxes, and Simple Interest

# Reminders and To-Do

- Quiz (Topic 2) at the beginning of class.
- Topic 3 will cover sections 6.2 and 6.3 of the text.
- Worksheet 3 at the end of class.
- Quiz (Topic 3) will be on Wednesday morning.

## Reminder

Your first **SC exam** is next Monday and will cover topics 1-4. Continue to review previous topics this week to take advantage of the second chance. Some notes:

- Remember there are no “quiz” or “exam” grades. You are only graded on each topic.
- The grade for a given topic is the highest score between that topic’s quiz and that topic’s portion on the SC exam.

# Percent Change

Sometimes it is helpful to measure a change in value as a **percent change**. For example, going from 10 to 20 is a 100% increase. Going from 10 to 5 is a 50% decrease.

## Definition

The **percent change** of a number  $n_i$  to a number  $n_f$  is

$$n_f = n_i(1 \pm p)$$

where  $p$  is the percent, written as a *decimal*. If the number went up (down), it is called a **percent increase (decrease)**.

You must decide when to use  $+$  (increase) and when to use  $-$  (decrease).

## Example: Percent Increase

A store buys their bags of flour from a wholesale vendor at a price of \$2 per bag. In order to turn a profit, they markup the price by 15%. What is the price they sell the flour for?

## Example: Percent Decrease

You go to the store and pick out a pair of shoes that cost \$30. You have a coupon for 20% off. How much do you pay?

## Example

The UConn bookstore buys a math textbook from the publisher for \$160. They markup the price by 97%. How much do you pay for the book?

## Example: Finding Percent Increase

If there are 250 students in a class in January, and 300 students in March, what is the percent change from January to March?



## Example: Decrease

If there are 250 students in a class in January, and 225 students in March, what is the percent change from January to March?

# Sales Tax

In most places, any purchase made is subject to **sales tax**.

## Interaction of Tax and Discounts/Markups

Since sales tax is applied to *purchases*, and coupons/discounts/markups are applied before you make the payment, sales tax takes effect **after** any discounts/markups.

## Example

You have a 20%-off coupon for a take-out place. Sales tax in this county is 6.25%. You want to buy a chicken sandwich for \$18.00.

- 1 What is the pre-tax total for the chicken sandwich?
- 2 What is the total you'll pay?

# Interest

A bank will use your deposited money to make more money for themselves, e.g. by investing it. You are entitled to a portion of those earnings.

## Definition

**Interest** is money paid for the privilege of being able to use someone else's money.

Note that interest goes both ways: if you take out a loan, you will owe the lender interest!

# Financial Terminology

## Definition

- ① The **principal** is the amount of money originally being deposited or loaned out (the base amount).
- ② The **interest rate** is the percentage used to determine an interest payment.
- ③ If an interest payment uses the interest rate with only the principal, then the account is said to be earning **simple interest**.

*Remark:* We will learn later that most interest is **compound**, which means the interest rate looks at the current value of the account (instead of the principal).

# Calculating Simple Interest

## Formula: Simple Interest Formula

If  $P$  is the principal,  $r$  the interest rate (written as a decimal), and  $t$  the time period in years, then the simple interest earned is given by

$$I = Prt.$$

## Example

An account has an initial deposit of \$500, paid 5% interest annually. How much interest has the account earned after 3 years? How much interest was earned after 3 months?

## Example

You deposit \$800 into a bank account with a simple interest rate of 3%.  
What is the amount of interest you earn after 6 months?

# Solving for the Interest Rate

Much like other problems we have worked on, we can use algebra to solve for other parts of the formula.

## Example

Let's say you deposited \$1000 into an account, and after 2 years, you've earned \$140 in simple interest. What is the interest rate?

# Future/Maturity Value

## Definition

The total amount of money you have (or owe) is called the **future value** (or **maturity value**):

$$A = P + I.$$

Since  $I = Prt$ , we have two other equivalent formulas for future/maturity value:

## Formula: Future/Maturity Value

$$A = P + Prt \quad \text{or} \quad A = P(1 + rt).$$



## Example

You deposit \$2 000 into a bank account with a 4% simple interest rate. What is the future value of the account after 18 months?

# Present Value

## How much money to start with?

A family wants to invest into an account earning 10% simple interest annually to pay for their newborn's future college. How much money should they invest *today* to have \$25 000 by the time their kid is 18?

## Definition

The amount of money needed to invest *now* in order to reach a prescribed future value is called the **present value**:

$$P = \frac{A}{1 + rt}$$

*Note:* Mathematically, present value and principal are the same.

# Loans and Partial Payments

Sometimes one has to take out a large loan and it may not be possible to pay it all off at once. One may opt to do **partial payments**, i.e. paying only part of the loan at a time.

When a partial payment is made, interest is prioritized. Any money remaining in the partial payment can then go towards the principal.

## Example

A loan of \$10 000 is taken out with a simple interest rate of 7%. After 30 days, a partial payment of \$1 000 is made.

- ① How much of the payment goes toward the interest?
- ② How much of the payment goes toward the principal?
- ③ What is the **remaining balance**, i.e. how much left you owe on the loan?